

# Transfer that actually transfers

Building age drives urban retrofit and energy policy, but records are incomplete. We classify the construction era of buildings within 30-metre Landsat pixels into four classes, training on Madrid and adapting to Amsterdam with as few as five labelled examples per class.

The organiser baseline showed substantial source-target mismatch, motivating a representation-based transfer strategy. Madrid learns feature ranking and class geometry; Amsterdam support labels estimate target prototypes. At five shots we shrink noisy prototypes toward re-centred Madrid geometry. At larger budgets shrinkage switches off while dimensionality reduction remains useful. In frozen EXP-010 audit episodes, tuned macro F1 improves over the matched full-feature, target-only control at every budget, with 45 of 50 paired wins. The largest mean gain occurs at five shots per class—only 20 labelled Amsterdam samples total—where prototype estimation is hardest.

# How old are these buildings?

- Four construction eras, per 30 m Landsat pixel
- Train on Madrid → adapt to Amsterdam
- As few as five labelled examples per class
- Metric: four-class macro F1

**76k**

Madrid pixels

**26k**

Amsterdam pixels

**42**

years of imagery

**5**

labels, worst case

# Construction leaves a spectral fingerprint

- Six spectral bands · once a year · 1984–2025
- A field becomes a roof. Concrete weathers.
- 42-year trajectory → 60 summary features
- Labels are area-weighted averages — noisy by construction

**One pixel can hold a 1950s block and a 2010s block — and gets a label describing neither.**

There is a ceiling below 1.0. Nobody knows where it is.

# Source-target mismatch motivates transfer

## Organiser reference path

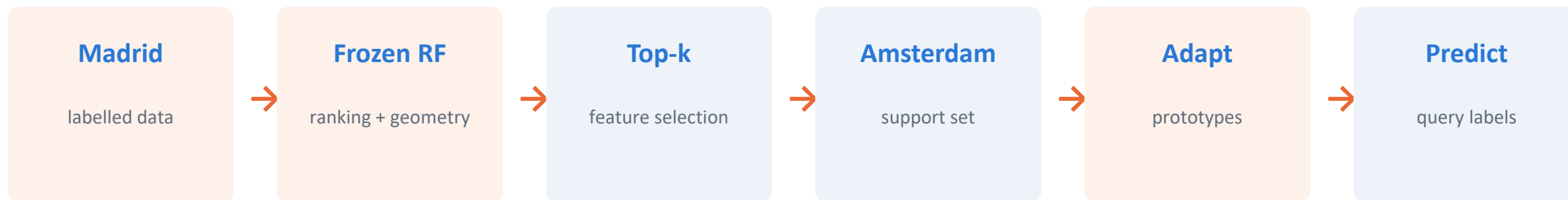
Madrid-fitted scaling

Amsterdam-only prototypes

- The reference path establishes a matched target-only control
- It motivated reuse of supervised Madrid state
- Our comparison holds episodes and classifier form fixed

**Provenance limitation: local RF outputs differ from organiser-saved outputs. Incomplete package/input provenance means the cause remains unidentified.**

# Freeze Madrid knowledge; adapt prototypes



**5 shots/class →  $k=30$ ,  $\lambda=0.6$**   
**Shrink toward Madrid geometry**

**25–200 shots/class →  $k=45$ ,  $\lambda=0.0$**   
**Shrinkage switches off**

Query labels are evaluation-only.

# Five shots is the practical anchor

| shots/class | EXP-010       | control       | gain   | wins  |
|-------------|---------------|---------------|--------|-------|
| 5           | 0.552 ± 0.060 | 0.515 ± 0.070 | +0.037 | 7/10  |
| 25          | 0.612 ± 0.013 | 0.609 ± 0.015 | +0.003 | 8/10  |
| 50          | 0.617 ± 0.008 | 0.614 ± 0.008 | +0.004 | 10/10 |
| 100         | 0.620 ± 0.005 | 0.615 ± 0.005 | +0.005 | 10/10 |
| 200         | 0.624 ± 0.004 | 0.619 ± 0.004 | +0.005 | 10/10 |

± population SD across 10 audit episodes — not confidence intervals

+0.0372

5-shot macro F1 gain

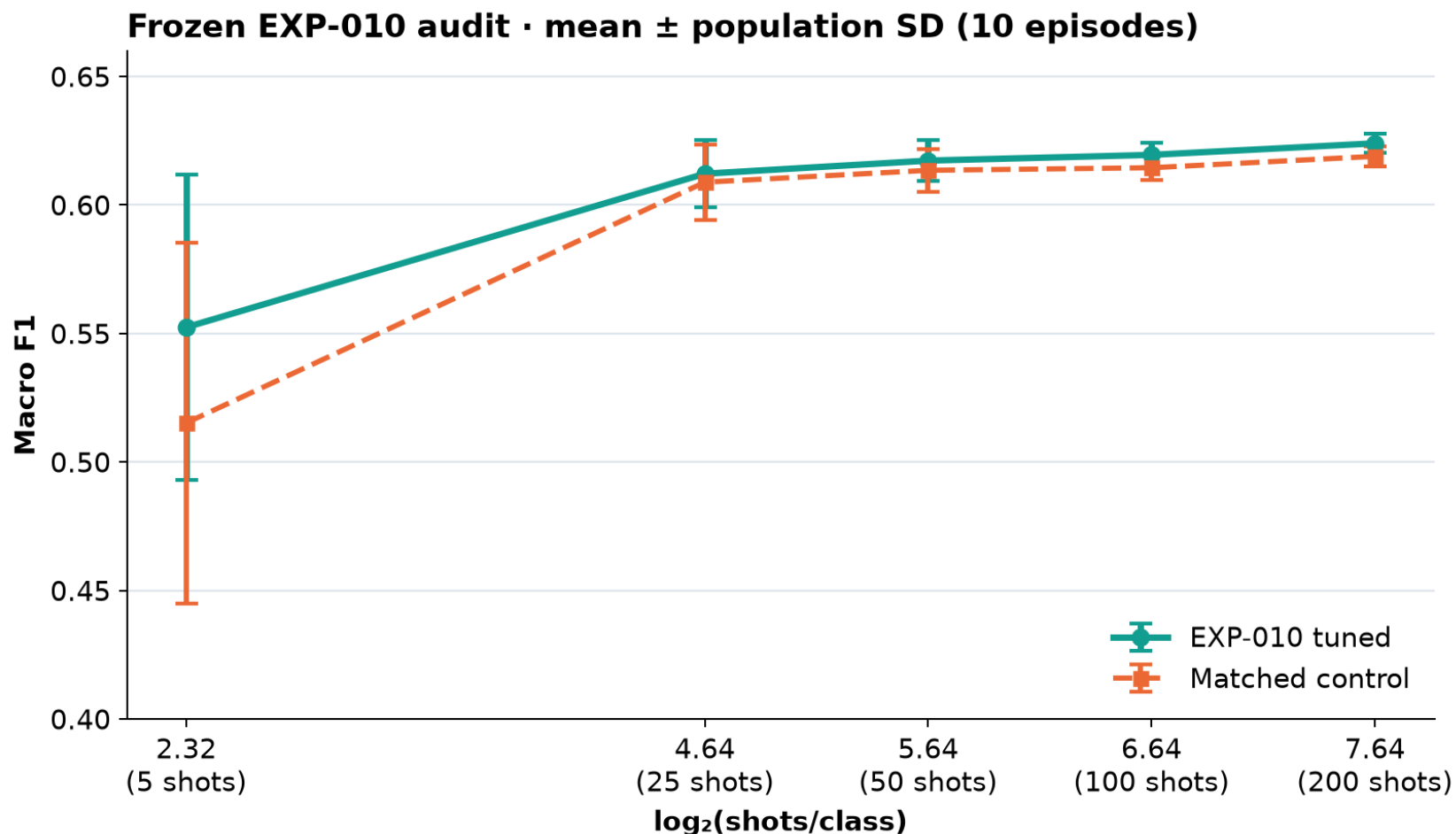
45 / 50

paired audit wins

SCIENTIFIC HEADLINE · 5/class = 20 labels total  
0.552 ± 0.060 vs 0.515 ± 0.070 · 7/10 wins

OFFICIAL 25-SHOT RESULT  
+0.0033 macro F1 · 8/10 paired wins

# Transfer helps most in the lowest-data regime



5 shots/class  
High episode spread

25+ shots/class  
Prototypes stabilise

**Empirical takeaway:**  
Transfer helps most when target  
prototypes are hardest to  
estimate.

Source-domain CV (documented local  
protocol):  
Madrid  $0.6281 \pm 0.0043$  macro F1

# More labels reduce prototype estimation error

| shots | features (k) | shrinkage ( $\lambda$ ) |
|-------|--------------|-------------------------|
| 5     | 30           | 0.6                     |
| 25    | 45           | 0.0                     |
| 50    | 45           | 0.0                     |
| 100   | 45           | 0.0                     |
| 200   | 45           | 0.0                     |

- $\lambda$  — how much we trust Madrid over Amsterdam
- k — how many features we keep
- Less data  $\rightarrow$  lean on Madrid, use fewer features
- More data  $\rightarrow$  trust Amsterdam, afford more features
- In these experiments, the selected schedule
- switches shrinkage off once labels increase.



# Negative results clarify the mechanism

## EXP-004 · WHITENING REVERSAL

Within-class whitening changed from harmful at low shot count to beneficial at high shot count. The same transform gave opposite outcomes as support grew.

## EXP-007 · MORE FEATURES DID NOT HELP

The expanded 94-feature seasonal/coverage representation did not outperform the original 60 features. Extra dimensions can dilute few-shot prototype estimates.

**Together, these outcomes support—without proving—the prototype estimation-error interpretation.**

# What we are not claiming

- Gains beyond five shots are small in absolute macro F1
- Error bars are spreads, not confidence intervals
- Confusion diagonals are recall, not F1
- Accuracy would have flattered us — we report macro F1
- We do not claim exact reproduction of organiser RF outputs

## Why not?

Source-domain Madrid CV is  $0.6281 \pm 0.0043$  under the documented local protocol; local zero-shot is 0.4433, versus organiser-saved 0.6179 and 0.3427. Package/input provenance is incomplete, so the cause remains unidentified and exact reproduction is not claimed.

# Limitations

1. Features still built label-dependently — no hidden-test path yet
2. Scaling fitted before cross-validation — not fold-safe
3.  $\lambda$  schedule: five values from five selection trials — overfit risk
4. Optimal  $k$  moves once (30→45) — weak evidence for a trend
5. EXP-010 has not been independently replicated

**No claims of worldwide portability. No claims of calibrated confidence.**

# Reproducible by design, limited by evidence

## EVALUATION CONTROLS

- Exact episode manifests + RNG binding
- Support/query isolation
- Frozen EXP-010 state
- Matched episodes for tuned and control
- Population SD over 10 audit episodes

## EVIDENCE BOUNDARY

- Madrid → Amsterdam only
- No hidden-test claim
- No independent EXP-010 replication
- Local baseline provenance is incomplete
- Controls support validity, not superiority

## CONCLUSION

**Madrid learns feature ranking and class geometry.**

**Amsterdam support labels estimate target prototypes.**

**EXP-010: positive mean gains, 45/50 paired wins.**

**A number without a mechanism is not a finding.**

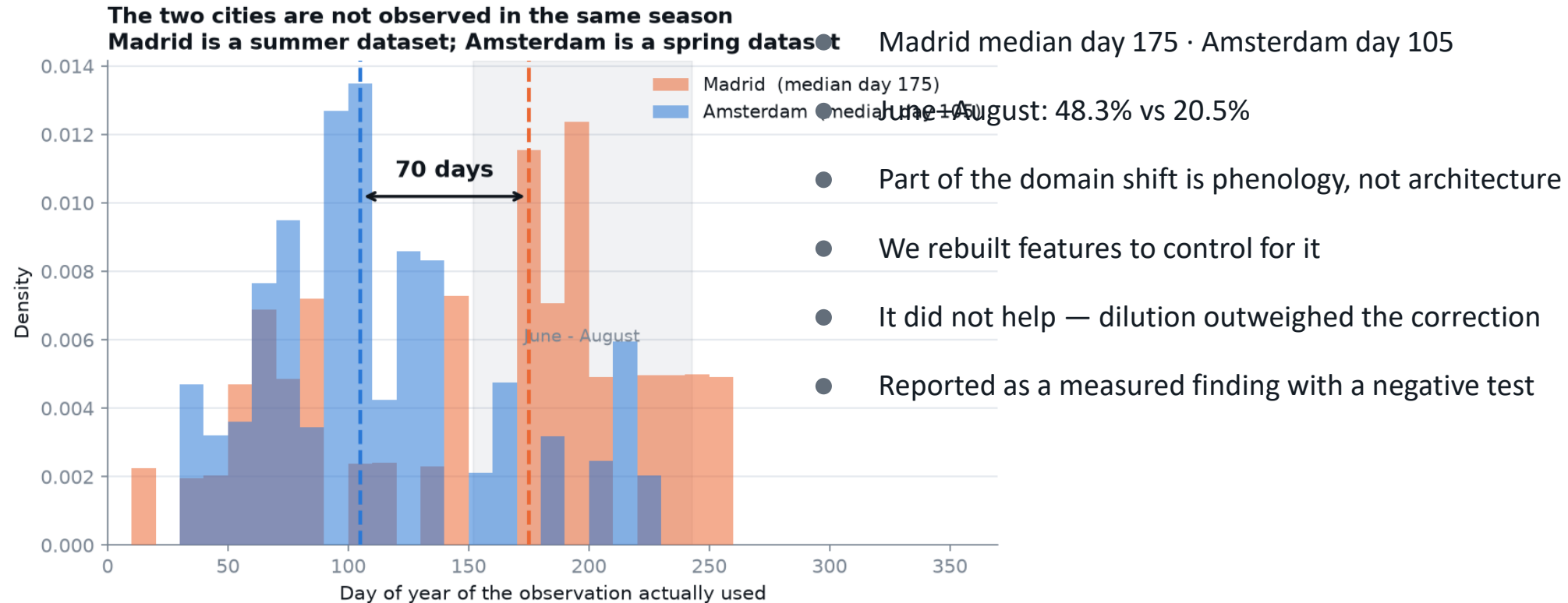
# Reproducibility

- requirements.txt pins the exact stack
- Raw data + feature cache SHA256 recorded
- Episode indices stored, not regenerated from a seed
- EXP-010 package stores weights, k=30/45 offsets and schedule
- Frozen artifacts include recorded SHA256 checksums
- Cold-start reproduction: ~35 minutes

# Why our baseline differs from the organiser's

- All six prototype budgets match them to 4 decimal places
- Both Random-Forest results differ: +0.0102 and +0.1006
- The prototype path does not consume fitted forest predictions
- Determinism check: 0 of 25,992 predictions differed across two refits
- Exact cause remains unidentified; no causal explanation is claimed

# A 70-day seasonal offset between the cities



June-August share: Madrid 48.3%, Amsterdam 20.5%. Part of the apparent domain shift is phenology, not architecture.



# The episode protocol

- One permutation per class per trial; budgets read prefixes
- So budgets are nested:  $\text{support}(5) \subset \text{support}(25) \subset \dots$
- Support and query disjoint, verified in code every episode
- Selection and audit sets never overlap
- Four independent seeds used across the project
- Query set shrinks unevenly as budget grows — a stated limitation